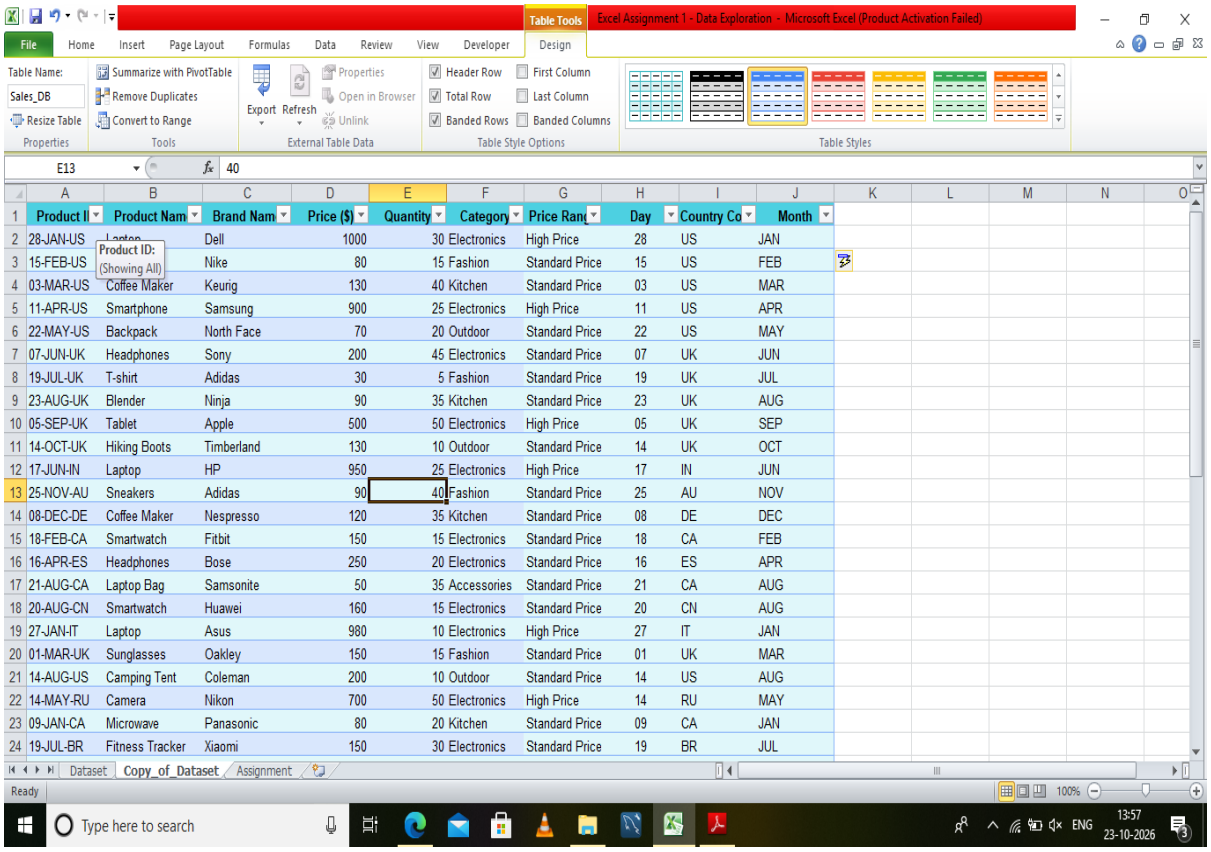
# Excel Assignment 1 – Data Exploration

## Tasks:

### *Basic Data Exploration*

Converting raw dataset into table and given Table Name as ***"Sales_DB".***




| Product I | Product Nam | Brand Nam | Price ($) | Quantity | Category | Price Ran | Day | Country Co | Month |
|---|---|---|---|---|---|---|---|---|---|
| 28-JAN-US | Laptop | Dell | 1000 | 30 | Electronics | High Price | 28 | US | JAN |
| 15-FEB-US | Sneakers | Nike | 80 | 15 | Fashion | Standard Price | 15 | US | FEB |
| 03-MAR-US | Coffee Maker | Keurig | 130 | 40 | Kitchen | Standard Price | 03 | US | MAR |
| 11-APR-US | Smartphone | Samsung | 900 | 25 | Electronics | High Price | 11 | US | APR |
| 22-MAY-US | Backpack | North Face | 70 | 20 | Outdoor | Standard Price | 22 | US | MAY |
| 07-JUN-UK | Headphones | Sony | 200 | 45 | Electronics | Standard Price | 07 | UK | JUN |
| 19-JUL-UK | T-shirt | Adidas | 30 | 5 | Fashion | Standard Price | 19 | UK | JUL |
| 23-AUG-UK | Blender | Ninja | 90 | 35 | Kitchen | Standard Price | 23 | UK | AUG |
| 05-SEP-UK | Tablet | Apple | 500 | 50 | Electronics | High Price | 05 | UK | SEP |
| 14-OCT-UK | Hiking Boots | Timberland | 130 | 10 | Outdoor | Standard Price | 14 | UK | OCT |
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| 21-AUG-CA | Laptop Bag | Samsonite | 50 | 35 | Accessories | Standard Price | 21 | CA | AUG |
| 20-AUG-CN | Smartwatch | Huawei | 160 | 15 | Electronics | Standard Price | 20 | CN | AUG |
| 27-JAN-IT | Laptop | Asus | 980 | 10 | Electronics | High Price | 27 | IT | JAN |
| 01-MAR-UK | Sunglasses | Oakley | 150 | 15 | Fashion | Standard Price | 01 | UK | MAR |
| 14-AUG-US | Camping Tent | Coleman | 200 | 10 | Outdoor | Standard Price | 14 | US | AUG |
| 14-MAY-RU | Camera | Nikon | 700 | 50 | Electronics | High Price | 14 | RU | MAY |
| 09-JAN-CA | Microwave | Panasonic | 80 | 20 | Kitchen | Standard Price | 09 | CA | JAN |
| 19-JUL-BR | Fitness Tracker | Xiaomi | 150 | 30 | Electronics | Standard Price | 19 | BR | JUL |

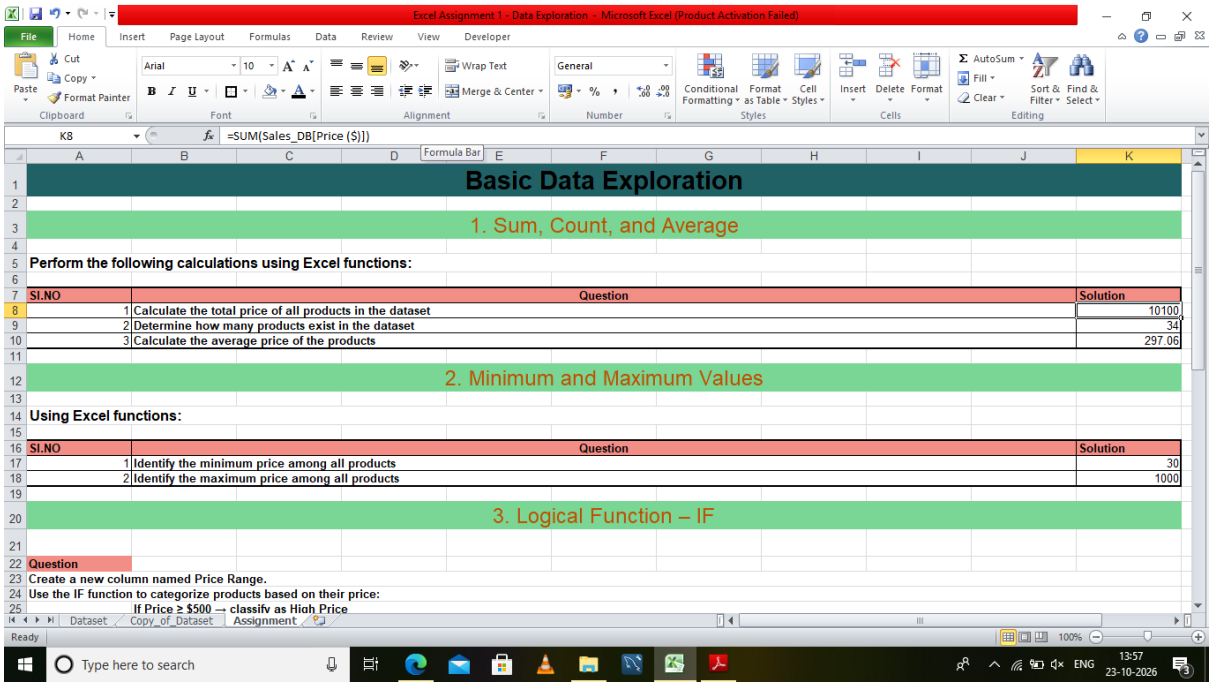
## Data Analytics (DA)

# Excel Assignment 1 – Data Exploration

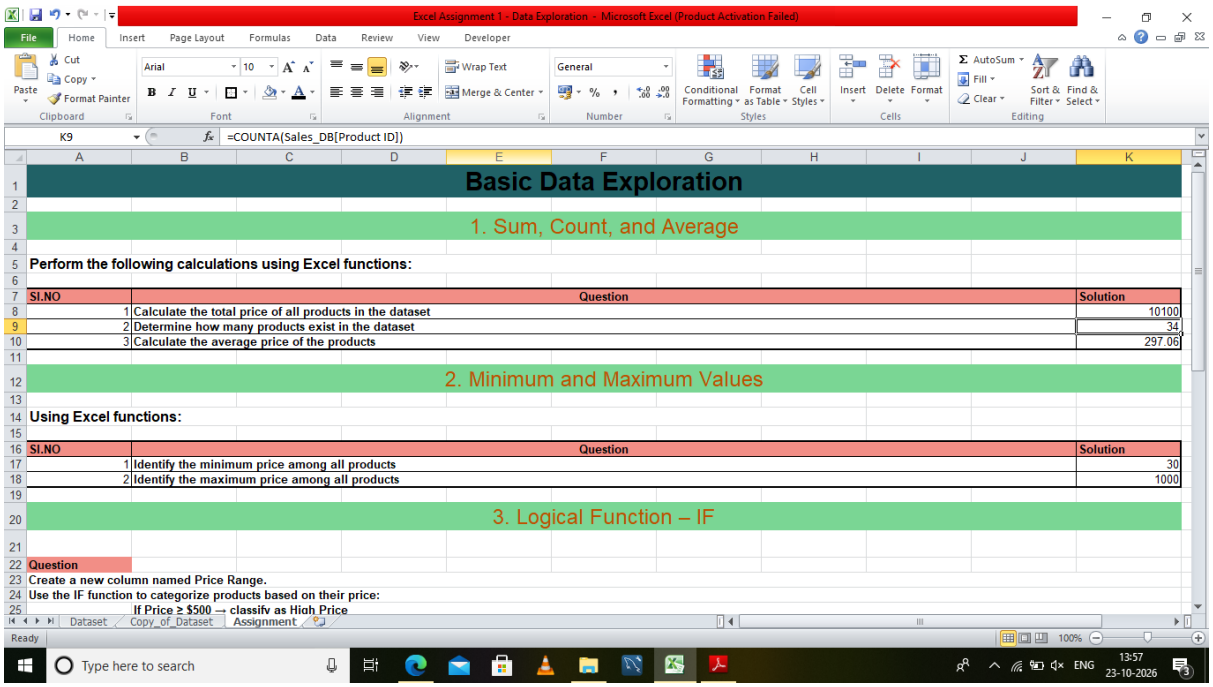
## 1. Sum, Count, and Average

Perform the following calculations using Excel functions:

- Calculate the **total price of all products** in the dataset.



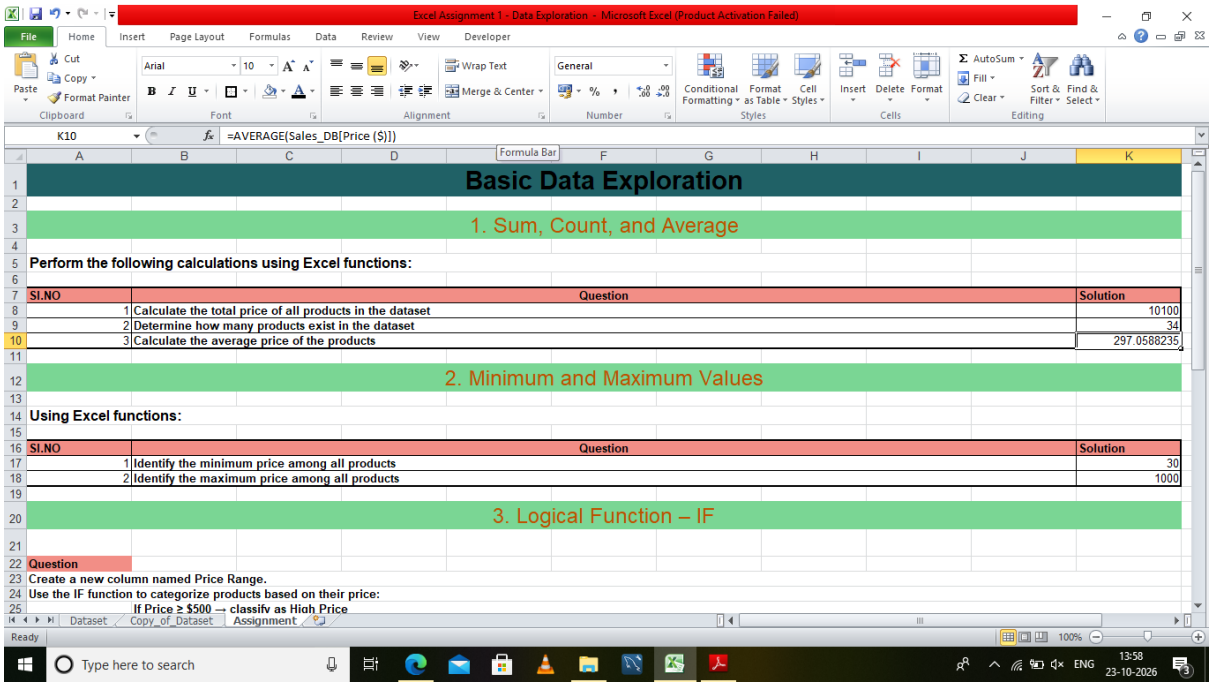

- Determine **how many products exist in the dataset.**




# Data Analytics (DA)

# Excel Assignment 1 – Data Exploration

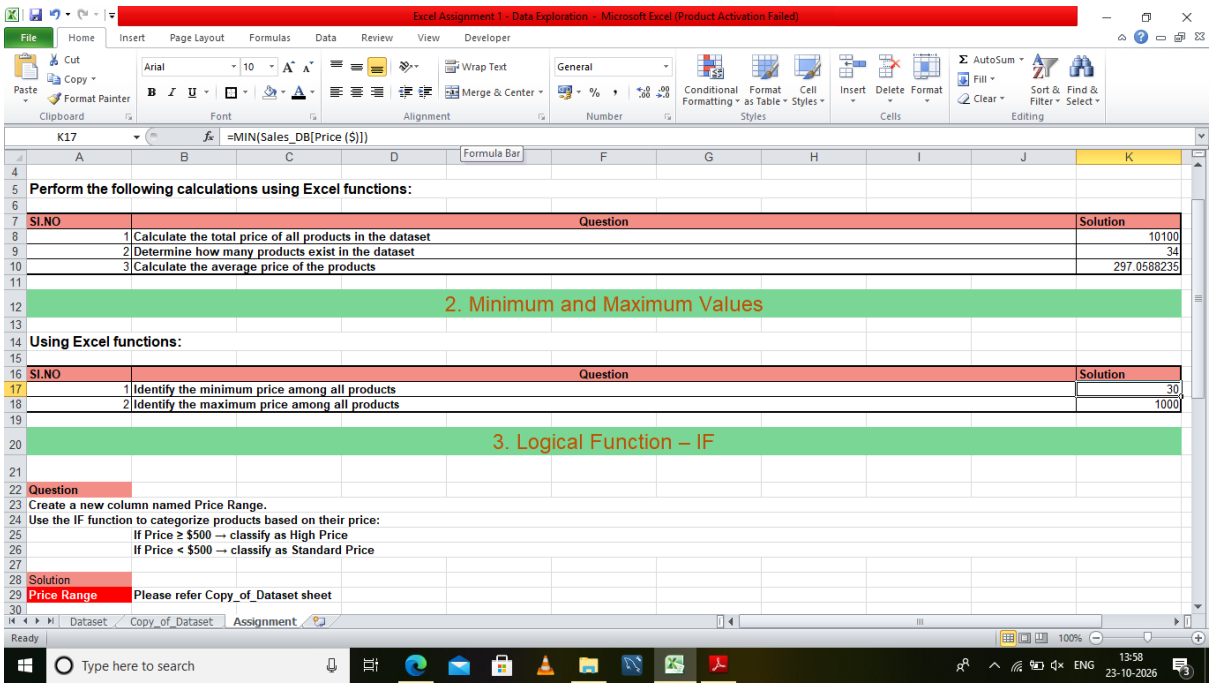
- Calculate the **average price of the products**.




## 2. Minimum and Maximum Values

Using Excel functions:

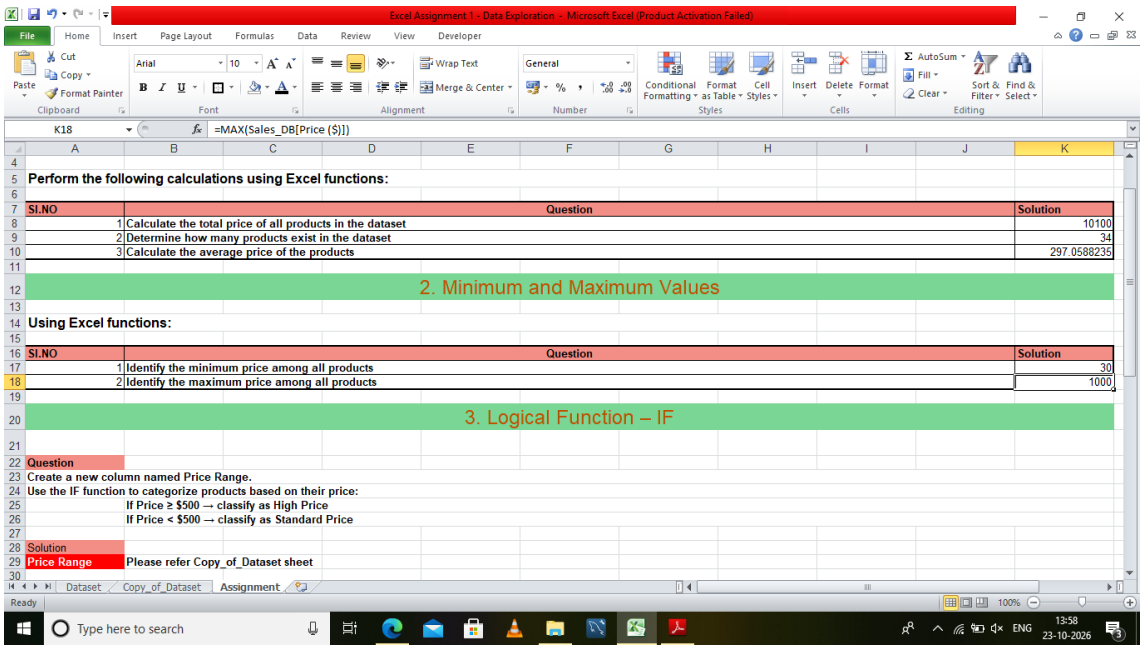
- Identify the **minimum price** among all products.




# Data Analytics (DA)

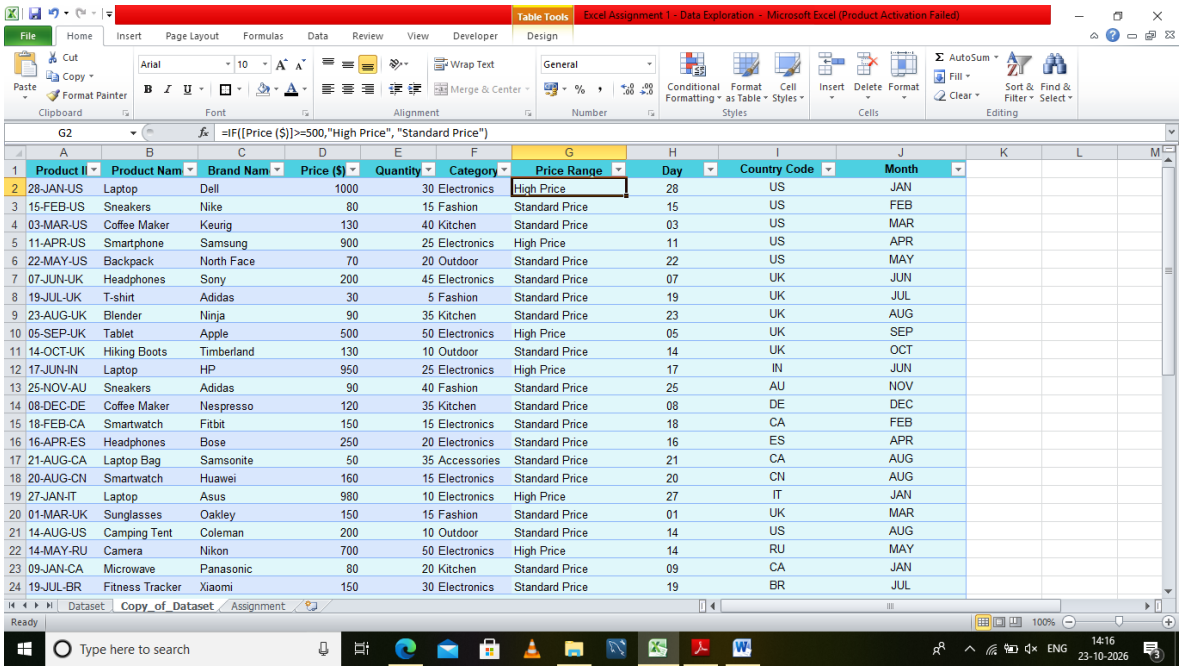
# Excel Assignment 1 – Data Exploration

- Identify the **maximum price** among all products.




## 3. Logical Function – IF

- Create a new column named **Price Range**.
- Use the **IF function** to categorize products based on their price:
    - If **Price ≥ $500** → classify as **High Price**
    - If **Price < $500** → classify as **Standard Price**



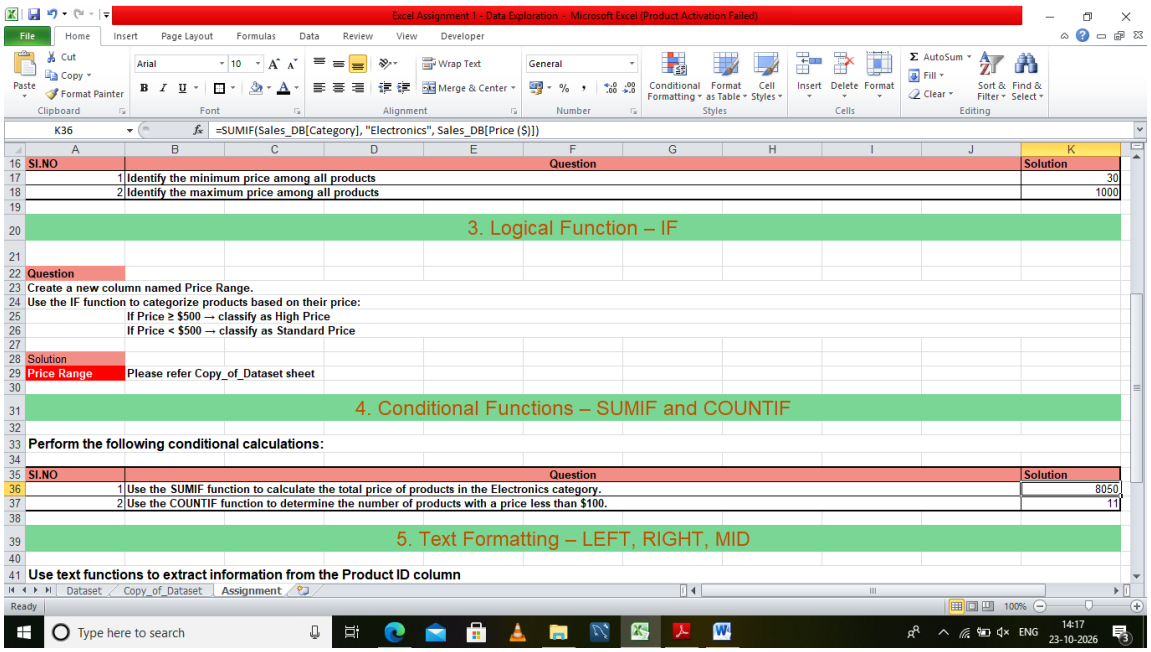

# Data Analytics (DA)

# Excel Assignment 1 – Data Exploration

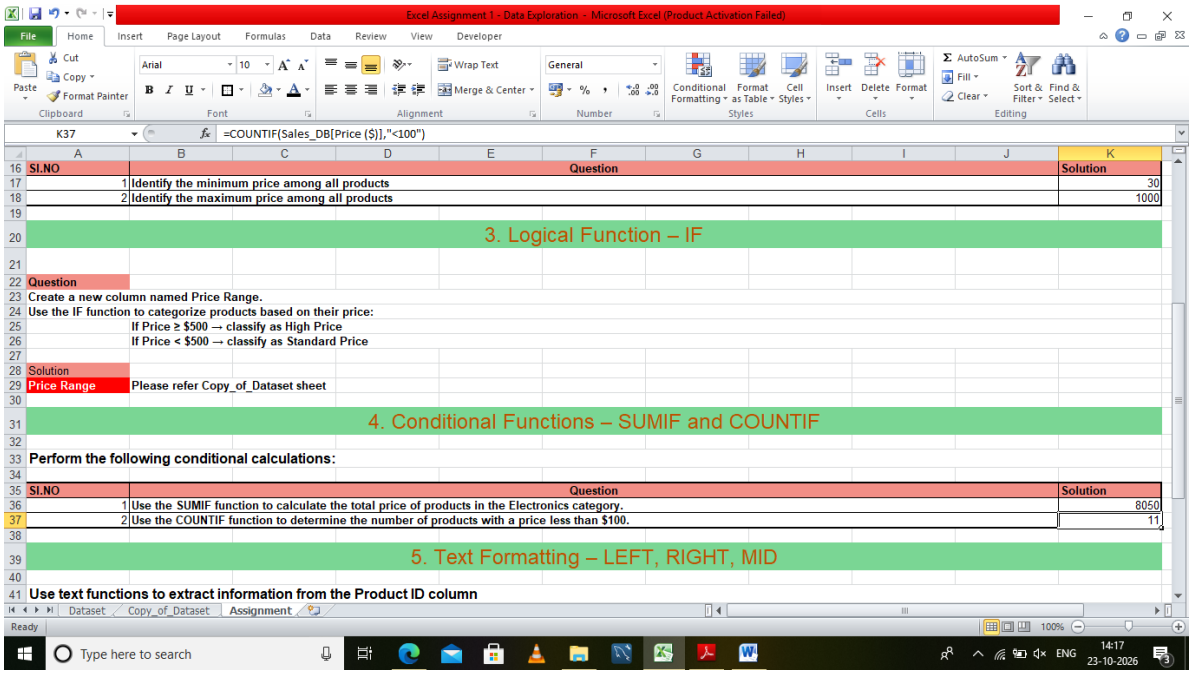
## 4. Conditional Functions – SUMIF and COUNTIF

Perform the following conditional calculations:

- Use the **SUMIF function** to calculate the **total price of products in the Electronics category**.




- Use the **COUNTIF function** to determine the **number of products with a price less than $100**.



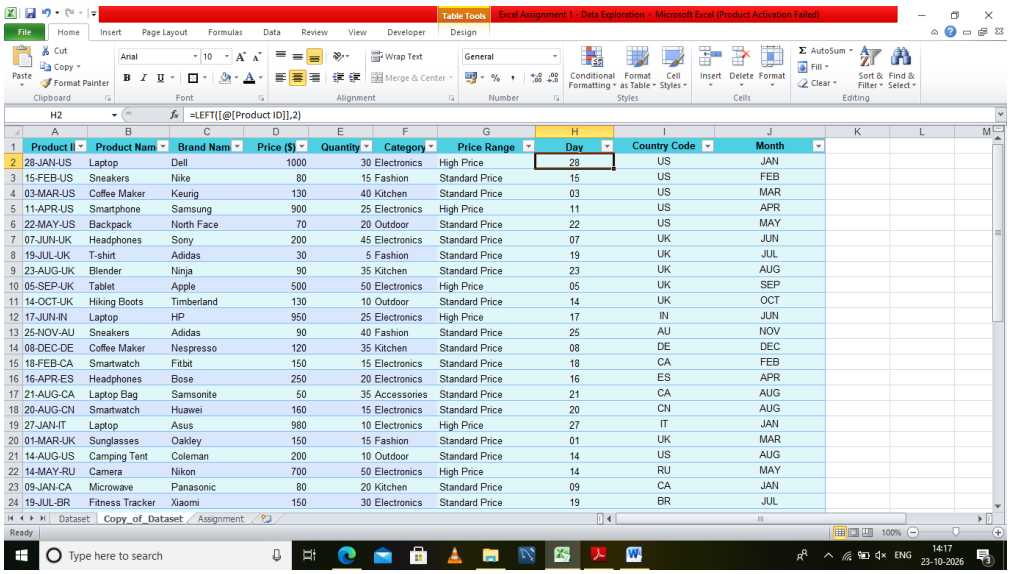

# Data Analytics (DA)

# Excel Assignment 1 – Data Exploration

## 5. Text Formatting – LEFT, RIGHT, MID

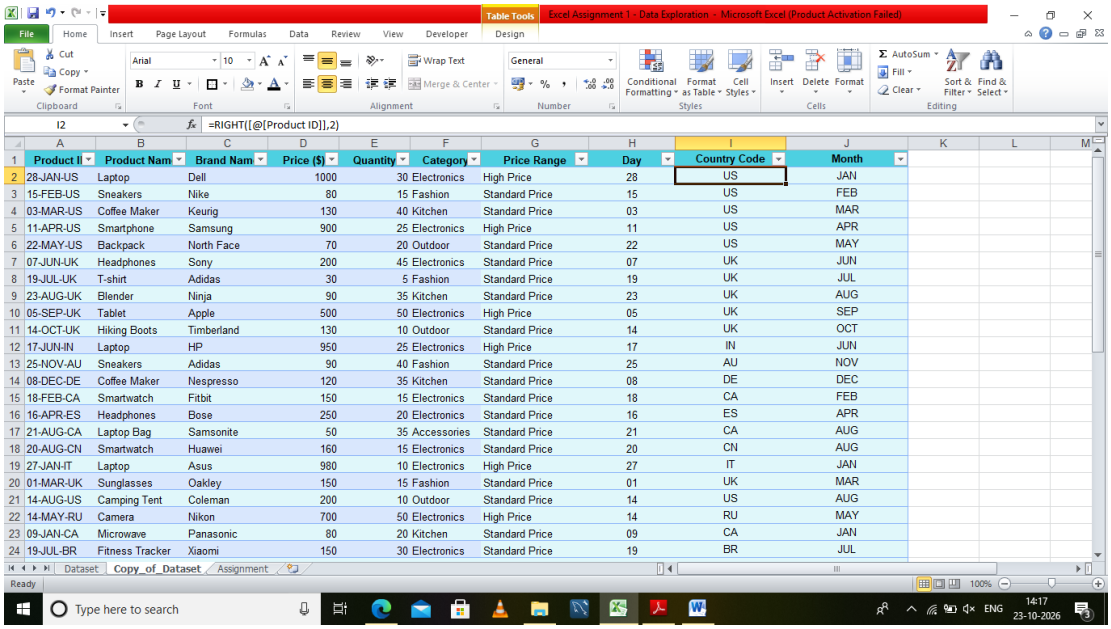
- Use text functions to extract information from the **Product ID** column. Create the following new columns:

○ Create a new column named **Day** with the first 2 characters of each **'Product ID'** using the **LEFT function**.




| Product ID | Product Name | Brand Name | Price ($) | Quantity | Category | Price Range | Day | Country Code | Month |
|---|---|---|---|---|---|---|---|---|---|
| 28-JAN-US | Laptop | Dell | 1000 | 30 | Electronics | High Price | 28 | US | JAN |
| 15-FEB-US | Sneakers | Nike | 80 | 15 | Fashion | Standard Price | 15 | US | FEB |
| 03-MAR-US | Coffee Maker | Keurig | 130 | 40 | Kitchen | Standard Price | 03 | US | MAR |
| 11-APR-US | Smartphone | Samsung | 900 | 25 | Electronics | High Price | 11 | US | APR |
| 22-MAY-US | Backpack | North Face | 70 | 20 | Outdoor | Standard Price | 22 | US | MAY |
| 07-JUN-UK | Headphones | Sony | 200 | 45 | Electronics | Standard Price | 07 | UK | JUN |
| 19-JUL-UK | T-shirt | Adidas | 30 | 5 | Fashion | Standard Price | 19 | UK | JUL |
| 23-AUG-UK | Blender | Ninja | 90 | 35 | Kitchen | Standard Price | 23 | UK | AUG |
| 05-SEP-UK | Tablet | Apple | 500 | 50 | Electronics | High Price | 05 | UK | SEP |
| 14-OCT-UK | Hiking Boots | Timberland | 130 | 10 | Outdoor | Standard Price | 14 | UK | OCT |
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| 18-FEB-CA | Smartwatch | Fitbit | 150 | 15 | Electronics | Standard Price | 18 | CA | FEB |
| 16-APR-ES | Headphones | Bose | 250 | 20 | Electronics | Standard Price | 16 | ES | APR |
| 21-AUG-CA | Laptop Bag | Samsonite | 50 | 35 | Accessories | Standard Price | 21 | CA | AUG |
| 20-AUG-CN | Smartwatch | Huawei | 160 | 15 | Electronics | Standard Price | 20 | CN | AUG |
| 27-JAN-IT | Laptop | Asus | 980 | 10 | Electronics | High Price | 27 | IT | JAN |
| 01-MAR-UK | Sunglasses | Oakley | 150 | 15 | Fashion | Standard Price | 01 | UK | MAR |
| 14-AUG-US | Camping Tent | Coleman | 200 | 10 | Outdoor | Standard Price | 14 | US | AUG |
| 14-MAY-RU | Camera | Nikon | 700 | 50 | Electronics | High Price | 14 | RU | MAY |
| 09-JAN-CA | Microwave | Panasonic | 80 | 20 | Kitchen | Standard Price | 09 | CA | JAN |
| 19-JUL-BR | Fitness Tracker | Xiaomi | 150 | 30 | Electronics | Standard Price | 19 | BR | JUL |

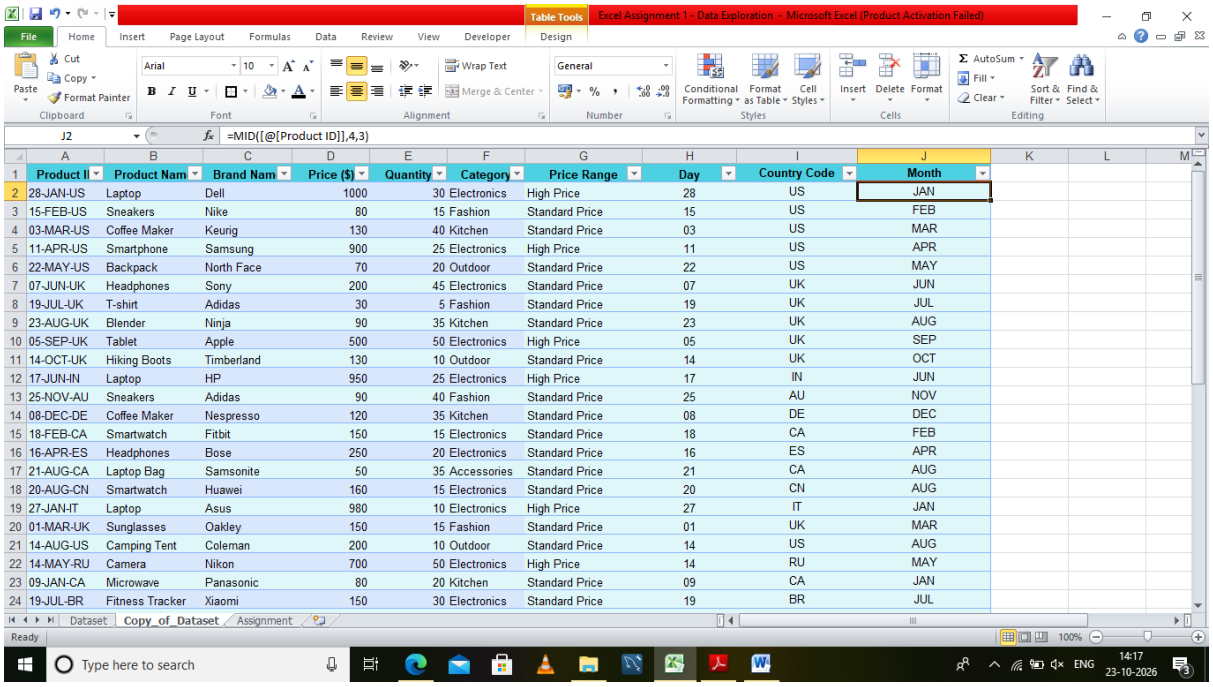
○ Create a new column named **Country Code** by extracting the last 2 characters from the **'Product ID'** column using the **RIGHT function**.




| Product ID | Product Name | Brand Name | Price ($) | Quantity | Category | Price Range | Day | Country Code | Month |
|---|---|---|---|---|---|---|---|---|---|
| 28-JAN-US | Laptop | Dell | 1000 | 30 | Electronics | High Price | 28 | US | JAN |
| 15-FEB-US | Sneakers | Nike | 80 | 15 | Fashion | Standard Price | 15 | US | FEB |
| 03-MAR-US | Coffee Maker | Keurig | 130 | 40 | Kitchen | Standard Price | 03 | US | MAR |
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| 22-MAY-US | Backpack | North Face | 70 | 20 | Outdoor | Standard Price | 22 | US | MAY |
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| 09-JAN-CA | Microwave | Panasonic | 80 | 20 | Kitchen | Standard Price | 09 | CA | JAN |
| 19-JUL-BR | Fitness Tracker | Xiaomi | 150 | 30 | Electronics | Standard Price | 19 | BR | JUL |

# Excel Assignment 1 – Data Exploration

○ Create a new column named **Month** by extracting the 4th to 6th characters from the **'Product ID'** column using the **MID function**.




| | Product Id | Product Nam | Brand Nam | Price ($) | Quantity | Category | Price Range | Day | Country Code | Month |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 28-JAN-US | Laptop | Dell | 1000 | 30 | Electronics | High Price | 28 | US | JAN |
| 3 | 15-FEB-US | Sneakers | Nike | 80 | 15 | Fashion | Standard Price | 15 | US | FEB |
| 4 | 03-MAR-US | Coffee Maker | Keurig | 130 | 40 | Kitchen | Standard Price | 03 | US | MAR |
| 5 | 11-APR-US | Smartphone | Samsung | 900 | 25 | Electronics | High Price | 11 | US | APR |
| 6 | 22-MAY-US | Backpack | North Face | 70 | 20 | Outdoor | Standard Price | 22 | US | MAY |
| 7 | 07-JUN-UK | Headphones | Sony | 200 | 45 | Electronics | Standard Price | 07 | UK | JUN |
| 8 | 19-JUL-UK | T-shirt | Adidas | 30 | 5 | Fashion | Standard Price | 19 | UK | JUL |
| 9 | 23-AUG-UK | Blender | Ninja | 90 | 35 | Kitchen | Standard Price | 23 | UK | AUG |
| 10 | 05-SEP-UK | Tablet | Apple | 500 | 50 | Electronics | High Price | 05 | UK | SEP |
| 11 | 14-OCT-UK | Hiking Boots | Timberland | 130 | 10 | Outdoor | Standard Price | 14 | UK | OCT |
| 12 | 17-JUN-IN | Laptop | HP | 950 | 25 | Electronics | High Price | 17 | IN | JUN |
| 13 | 25-NOV-AU | Sneakers | Adidas | 90 | 40 | Fashion | Standard Price | 25 | AU | NOV |
| 14 | 08-DEC-DE | Coffee Maker | Nespresso | 120 | 35 | Kitchen | Standard Price | 08 | DE | DEC |
| 15 | 18-FEB-CA | Smartwatch | Fitbit | 150 | 15 | Electronics | Standard Price | 18 | CA | FEB |
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